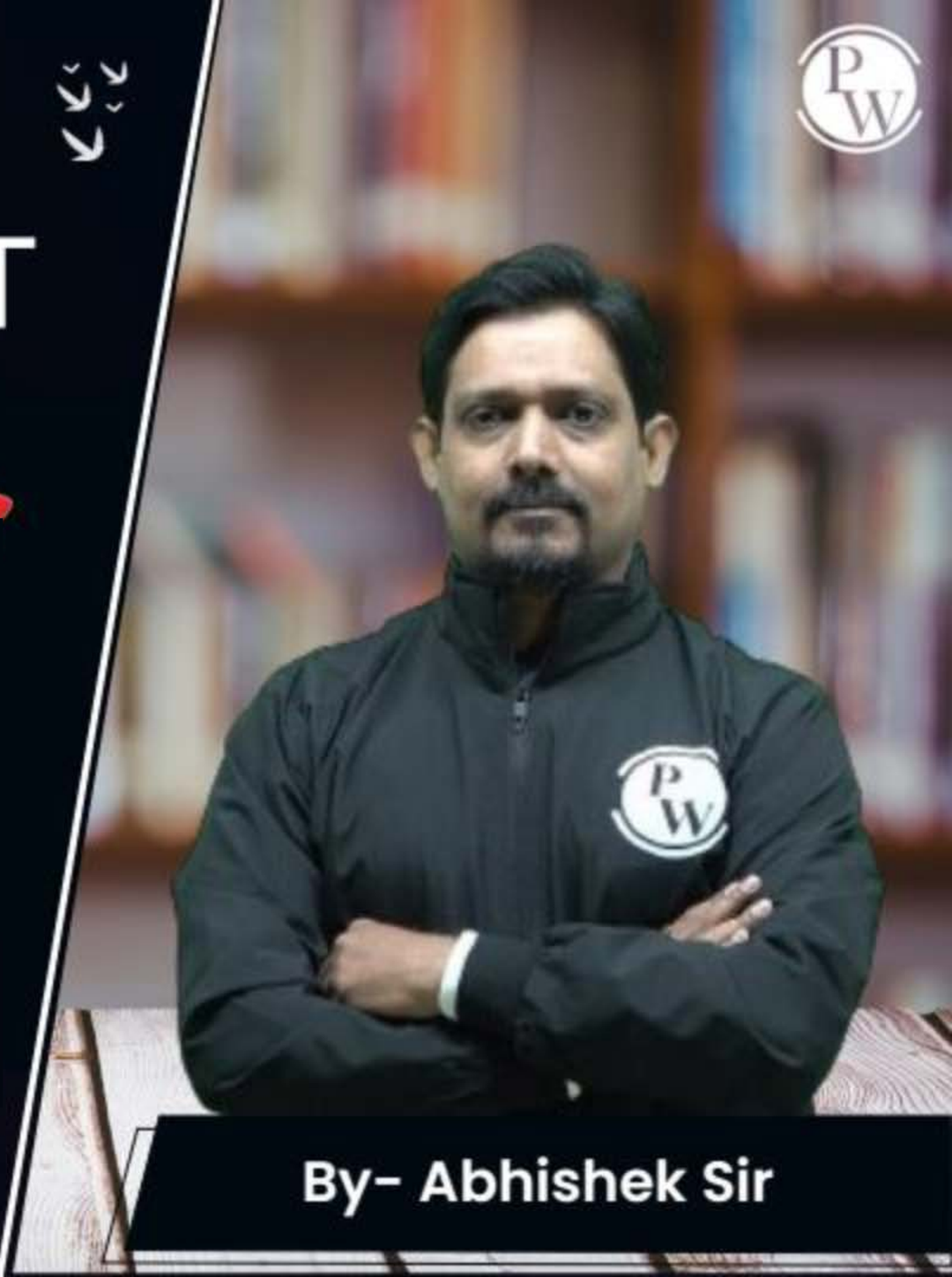
Computer Science & IT



C Programming

Functions & storage classes

Lecture No. 06



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Practice problem
- Topic — Recursion
- Topic
- Topic
- Topic

Slide

Topics to be Covered



Topic Extern & Register

Topic Tower of Hanoi

Topic

Topic

Topic

Slide

Auto : Automatic Destroy (Stack)

Defaul Local

Run time stack.

Static : Default global } Static data

Local static }

Initialize one time, assign every time

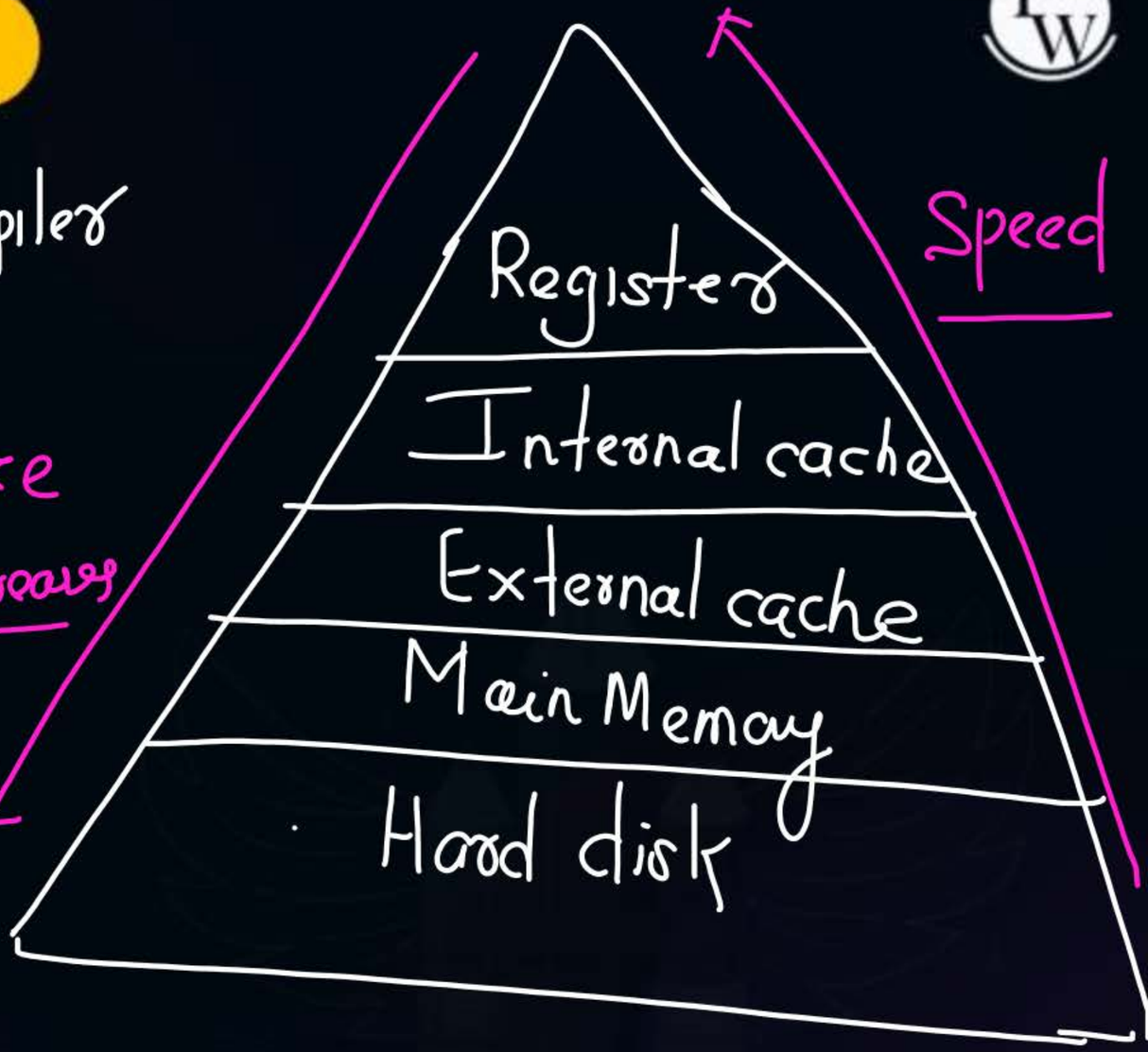


Register Storage class

Register allocation : Compiler

Register are inside CPU (processor)

Size Increases



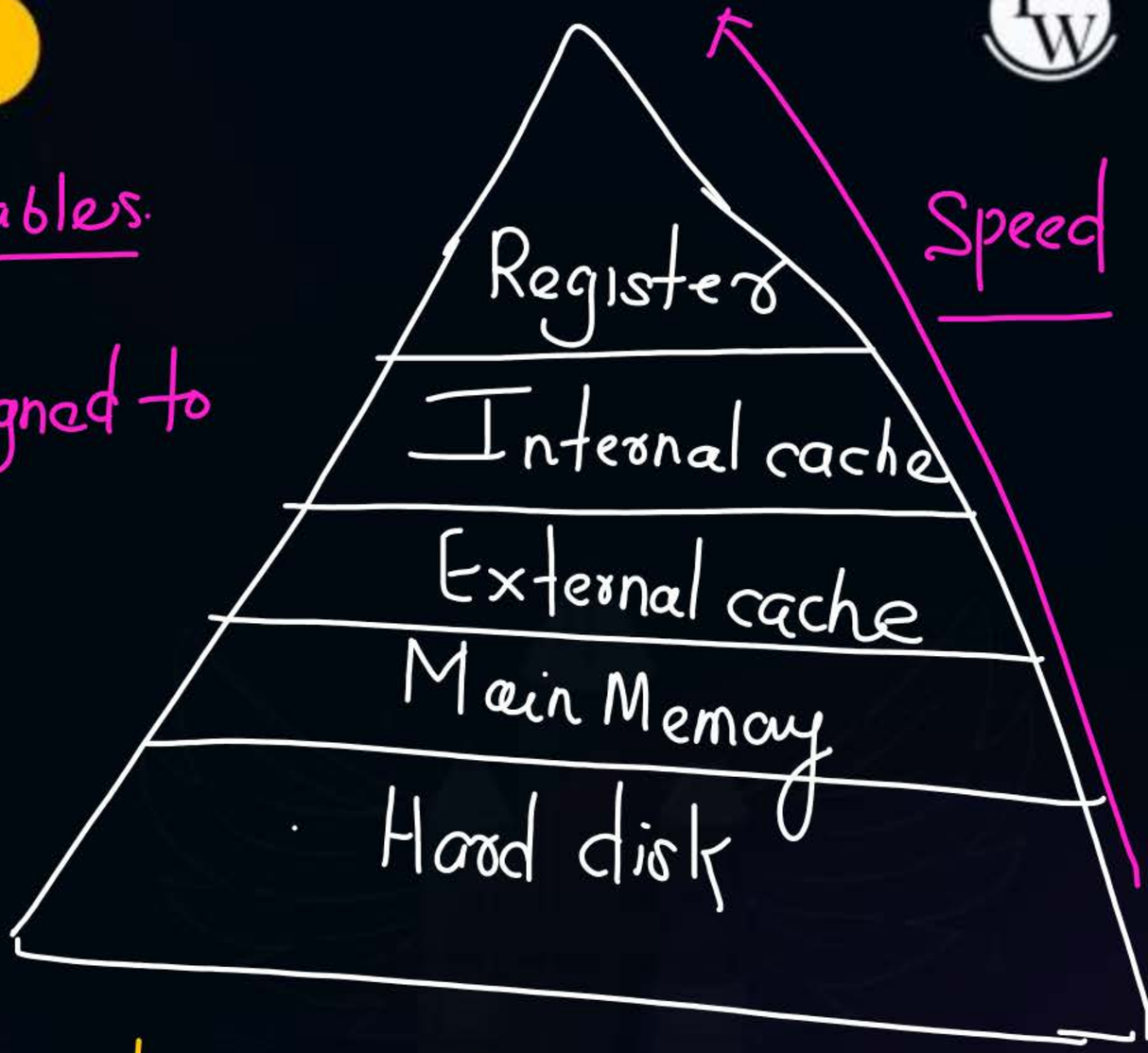
Slide

Register Storage class

program having Lot of variables.

Every variable can't be assigned to Register.

if in a program there is a Variable which is frequently access. We can Instruct compiler to assign Register to it



Slide

register int count;

⟸ Compiler is instructed to allocate Register to the variable.

frequently access variable

- GPRS
- general purpose Register
- local variable



Extern Storage Class

every variable is associated with Scope (visibility, accessibility, Lifetime)

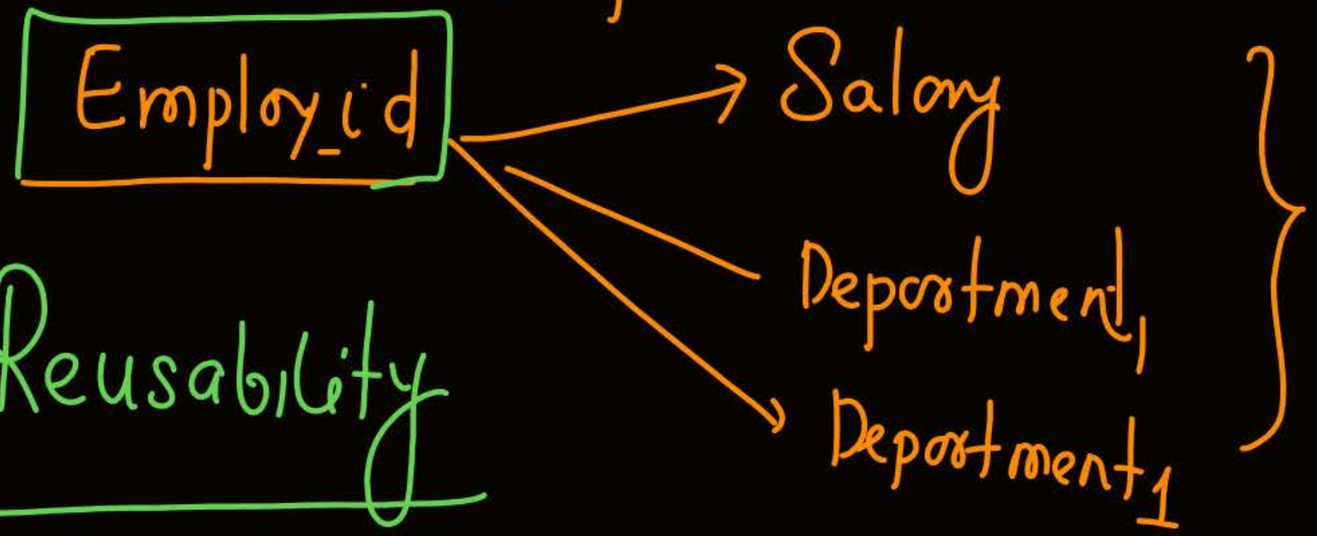
{ } block

Extern keyword extends visibility of a variable

Slide

What is the advantage

when big projects are developed then many file are present.



Every file/module has to declare variable employ_id.

Reusability



Extern Storage Class

you create one file that contains all the variables and let the other file Access it.

* To access other file data you need to extend the visibility by using extern keyword.
* Extern keyword applies on global variable.

Slide

Extern Storage Class

project
main.c
other.c

main.c

extern int a;

↑

Declaration

↓

No memory allocate

printf( )

other.c

int a = 5;

global variable.

Slide



Extern Storage Class

declaration

int a

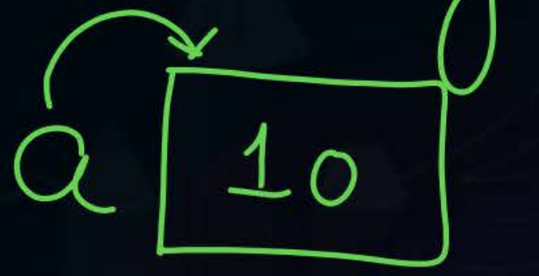
Associating Name to data type

int a = 10

a is int

definition

Associating a variable name to memory



Slide



project

main.c

other.c

```
extern int a;
int main(){
  printf("%d", a);
}
```

other.c

```
int a=5;
```

Slide



```
int a = 10;
int main(){
    extern int a;
    printf("%d", a);
}
```

→ 10

Visibility & Scope of variable a will be extended

Slide



Tower of Hanoi

* Given n disc of different size and 3 tower are given. (L, M, R)

* n disc are arranged in tower L from Largest to smaller and Largest disc in the bottom.

Slide



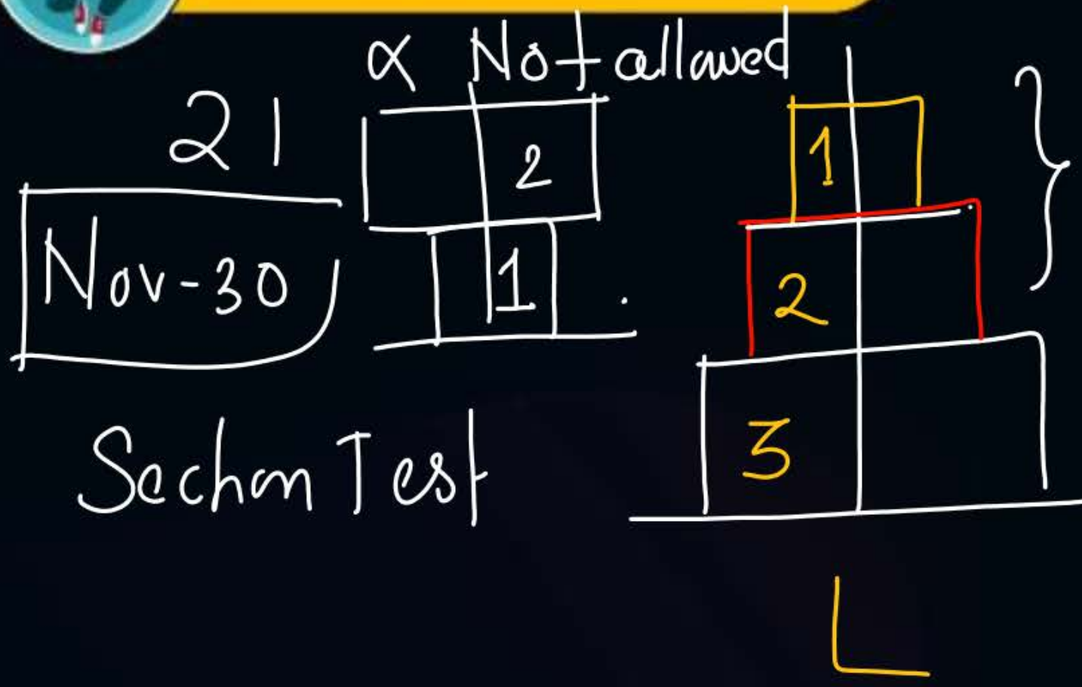
Tower of Hanoi

3 Question 100 / 50 days
Hrs

∝ Not allowed

21
Nov-30

Sachan Test



We Need to move 3 disk from tower L to tower R

* No large disc will be on top of smaller disc
* only one disc will be move at time.

Slide

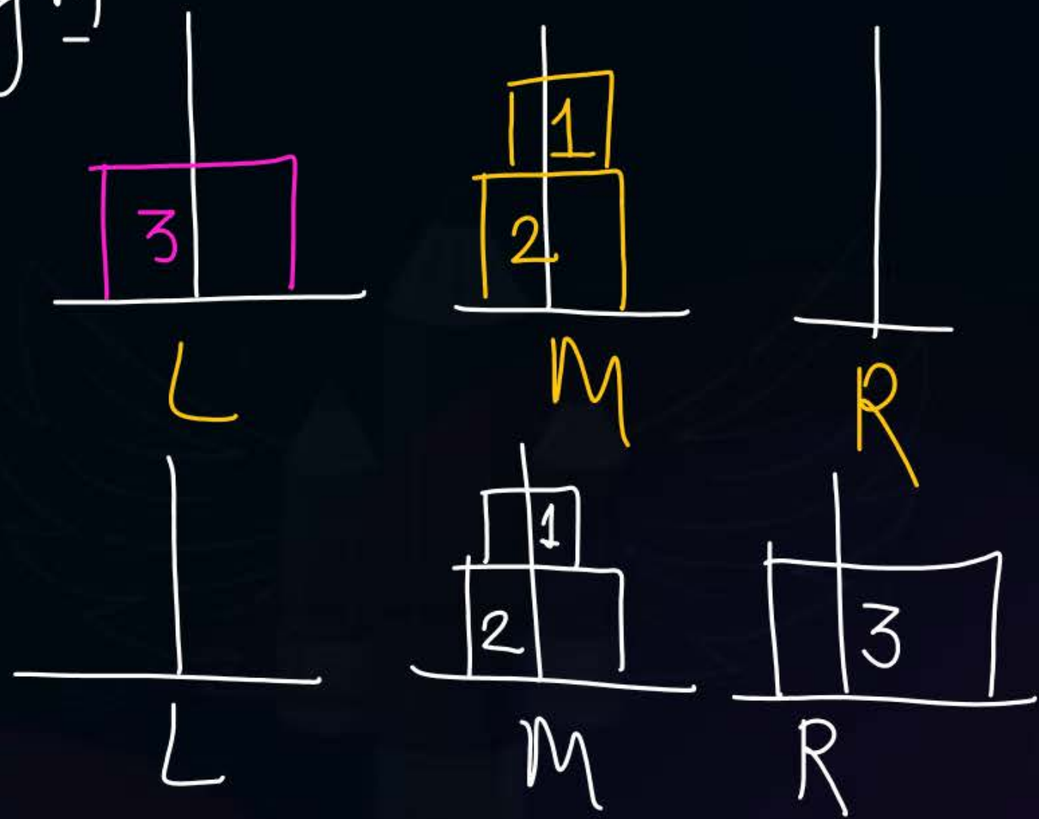


Tower of Hanoi

TOH(L, M, R, 3)

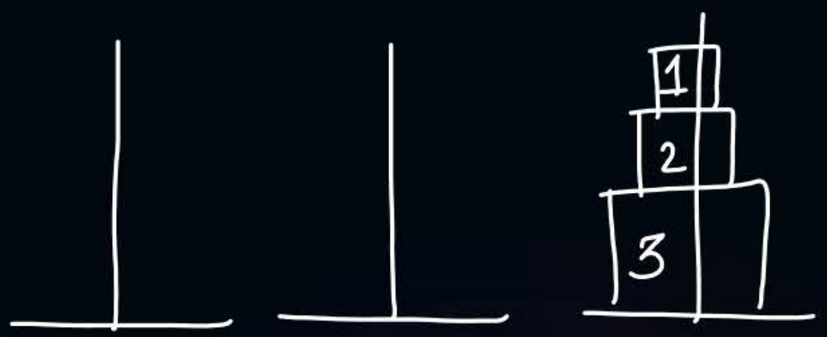
→ Move 3 disc from tower L to tower R using M

```
TOH(L, R, M, 2);
move(3rd disk L→R);
TOH(M, L, R, 2)
}
```



Slide

Tower of Hanoi




```
Algorithm TOH(L, M, R, n) {
    if(n>=1) {
        TOH(L, R, M, n-1)
        Move nth disc L → R
        TOH(M, L, R, n-1)
    }
}
```

Slide



Tower of Hanoi

TOH(L, M, R, 3)
1 2 3

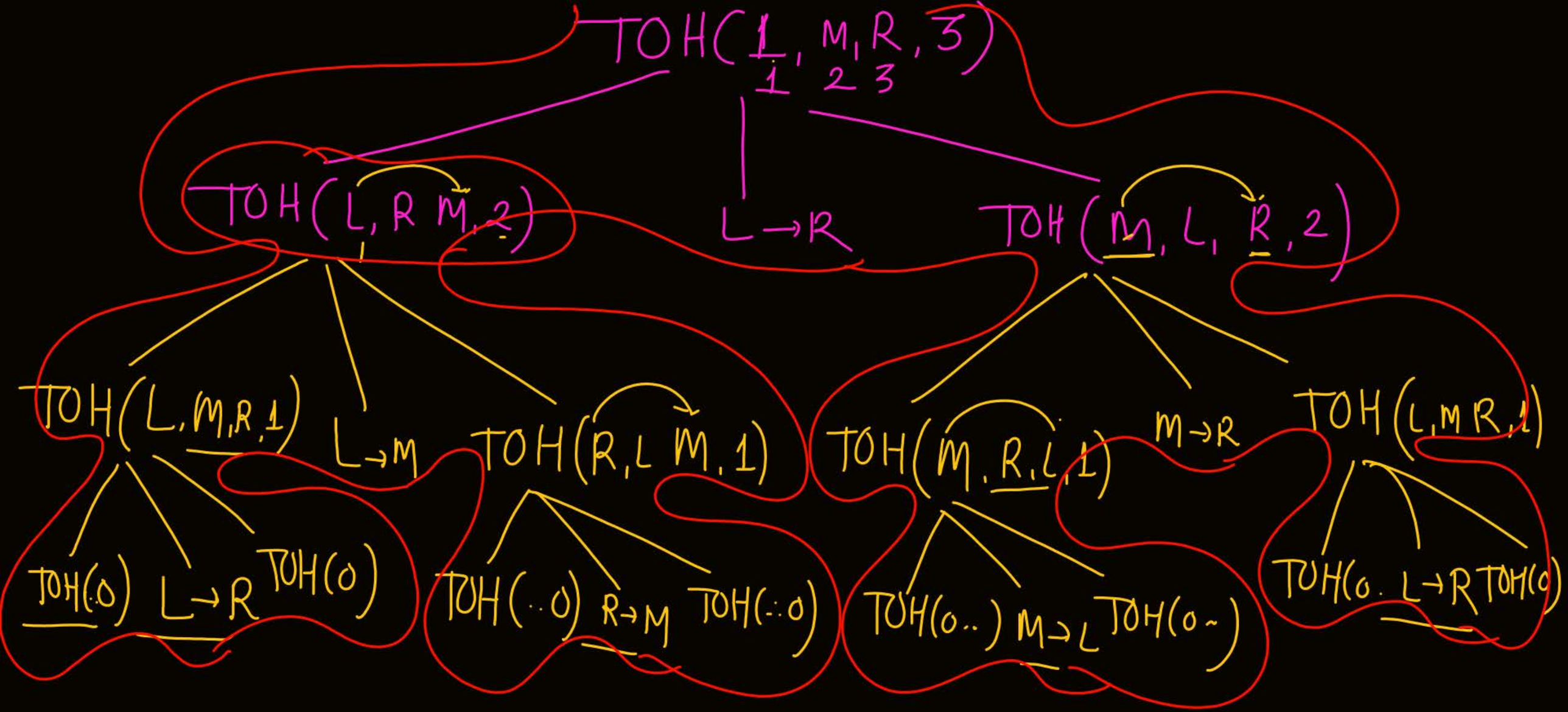
move 3 disk from L to R using M

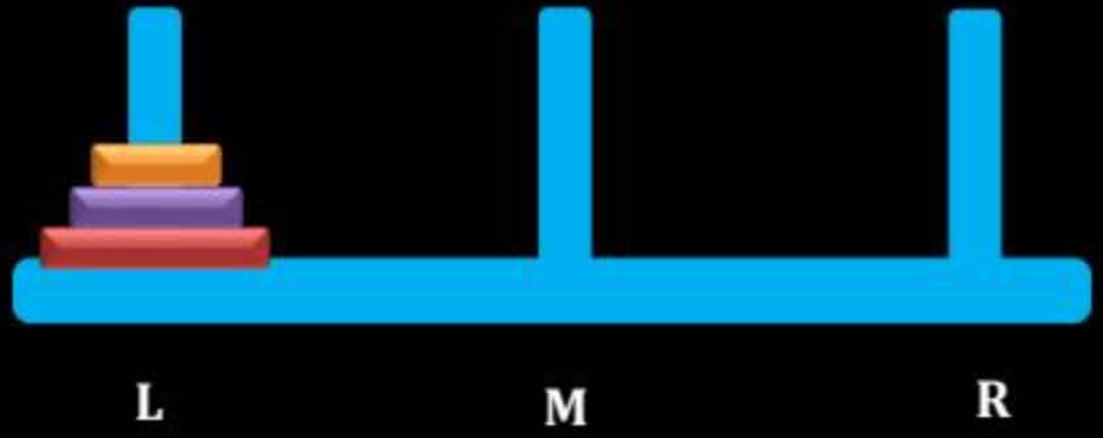
for disk movement

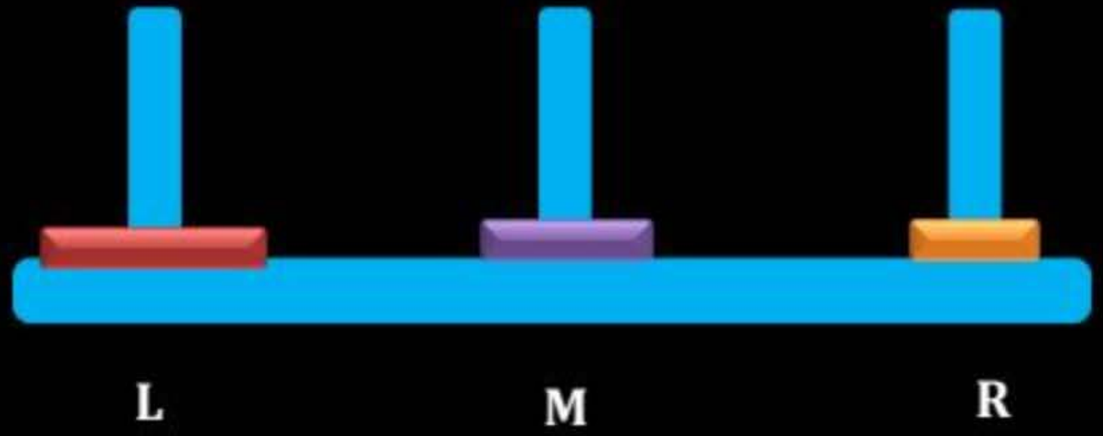
at least 1 disc

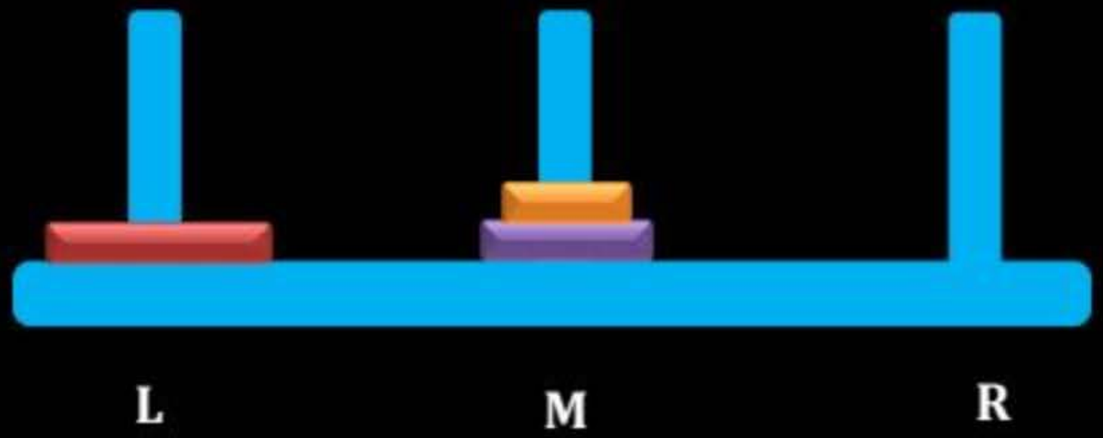
```
if(n>0){
① TOH(L, R, M, 2)
              1 3 2
② L → R
③ TOH(M L R, 2)
      2 1 3
}
```

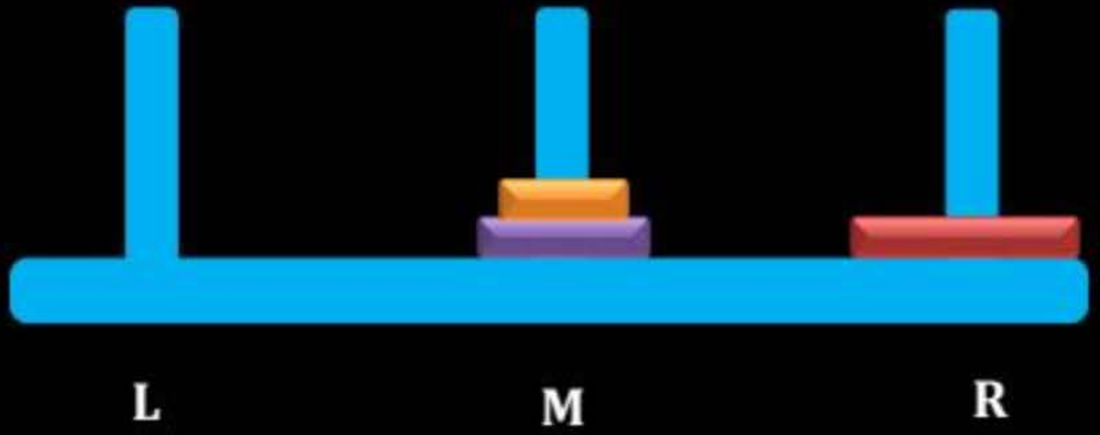
Slide

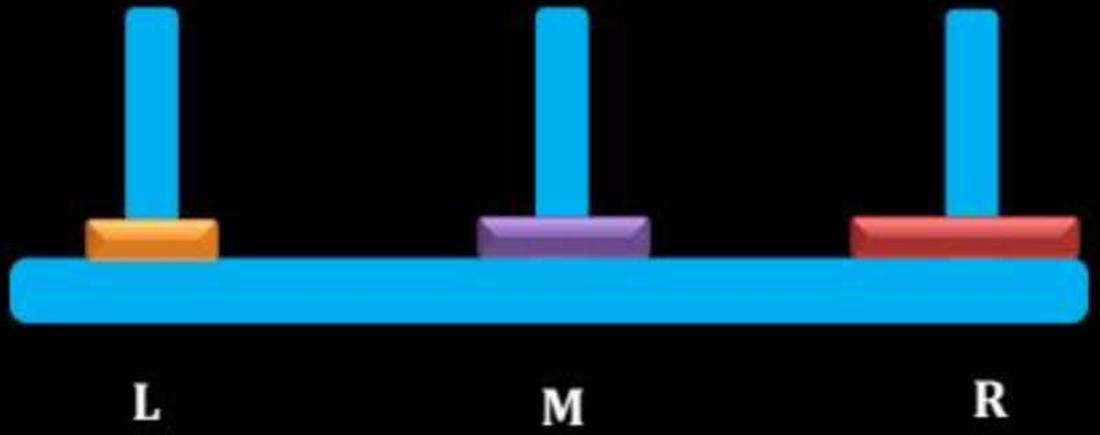

TOH(1, L, M, R, 3)
1 2 3
TOH(L, R, M, 2)
L→R
TOH(M, L, R, 2)
TOH(L, M, R, 1)
L→M
TOH(R, L, M, 1)
M→R
TOH(M, R, L, 1)
TOH(L, M, R, 1)
TOH(0)
L→R
TOH(0)
TOH(.0)
R→M
TOH(..0)
TOH(0..)
M→L
TOH(0~)
TOH(0.
L→R
TOH(0)

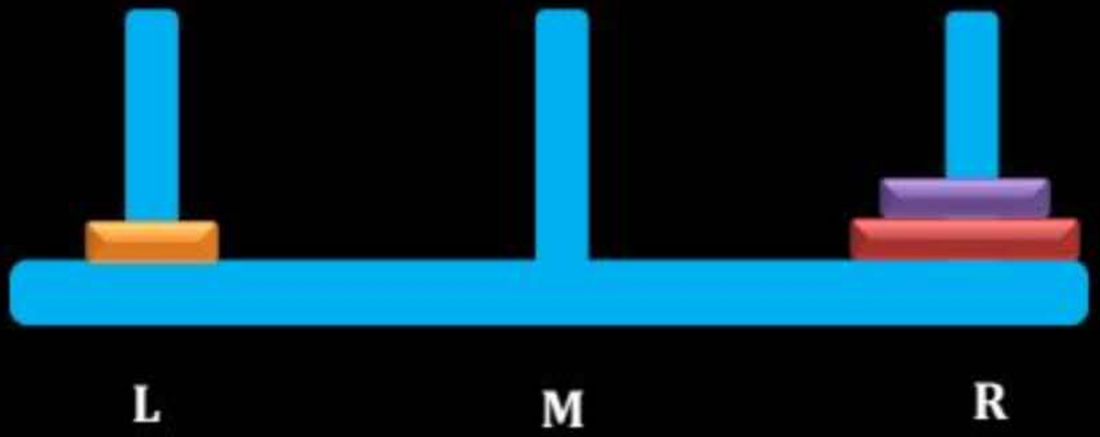

L
M
R


L
M
R


L
M
R


L
M
R


L
M
R


L
M
R



Pointer

pointer & array.

Slide



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Register

External.

TOH (Tower of Hanoi)

Slide

THANK - YOU

